

The exceptional C_3 orbital-measure base pair is singular

Full answer to an open base case in arXiv:2110.06346

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Status. Candidate full negative answer to the explicitly undecided C_3 base case, with a sharp codimension statement. This is a new-result claim subject to human review.

Source question

Hare's paper asks whether the exceptional pair

$$(C_2 \times C_1, C_1 \times C_1) \quad \text{in type } C_3$$

is absolutely continuous. This is the base case for one of the exceptional families that become entwined under the proposed rank-reduction procedure. The question appears on source PDF page 18.

(i) x is type $C_{n/2} \times C_{n/2}$ (or $C_{n/2} \times C_{n/2}$ (if even)) and y is the same type as x or type $SU(n)$;

(ii) x is type $C_{n/2} \times C_{n/2}$ and y is type $C_{n/2} \times C_{n/2-1}$ (n even) or x is type $C_{(n+1)/2} \times C_{(n-1)/2}$ and y is type $C_{(n-1)/2} \times C_{(n-1)/2}$ (n odd);

(iii) (in D_n only) x is type $SU(n)$ and y is type $SU(n)$ or $SU(n-1)$.

The pairs of (i) are eligible, but their reductions (defined in a similar manner as in B_n) are not. In fact, it can be seen from [6, Thm. 9.1] that the pair of type $(C_{n/2} \times C_{n/2}, C_{n/2} \times C_{n/2})$ is not absolutely continuous (although it is eligible). The pairs of (ii) are entwined under the reduction process and it is unclear whether their base case, $(C_2 \times C_1, C_1 \times C_1)$ in C_3 , is an absolutely continuous pair. The pairs in (iii) coincide with exceptional pairs from the Lie algebra problem, whose absolute continuity was undetermined in [7].

All non-exceptional, eligible tuples reduce to non-exceptional, eligible tuples.

Conjecture 5.9. Suppose the non-central elements x_j belong to the torus of C_n or D_n for $n \geq 5$ and that $(x_1, \dots, x_L)$ is not exceptional. Then $(x_1, \dots, x_L)$ is absolutely continuous if and only if it is eligible.

It is natural to appeal to an induction argument on the rank of the Lie group.

Set-up

Use the compact simply connected model

$$G = \mathrm{Sp}(3) = \{U \in M_3(\mathbb{H}) : U^*U = I\}, \quad \mathfrak{g} = \mathfrak{sp}(3) = \{A : A^* = -A\}.$$

Matrices act on the right quaternionic Hilbert space $\mathbb{H}^3$. For $z \in G$, write $\mathfrak{z}_z$ for its Lie centralizer and $\mathcal{N}_z = \mathfrak{z}_z^\perp$.

We may take

$$x = \mathrm{diag}(1, 1, -1), \quad y = \mathrm{diag}(1, -1, e^{i\theta}), \quad 0 < \theta < \pi.$$

Then x has type $C_2 \times C_1$, while y has type $C_1 \times C_1$; the omitted third factor is the empty root system $SU(1)$, whose centralizer contributes $U(1)$. Thus

$$\begin{aligned} \mathfrak{z}_x &\cong \mathfrak{sp}(2) \oplus \mathfrak{sp}(1), \\ \mathfrak{z}_y &= \{\mathrm{diag}(a_1, a_2, a_3) : a_1, a_2 \in \mathrm{Im} \mathbb{H}, a_3 \in \mathbb{R}i\}. \end{aligned}$$

The geometric criterion in source Proposition 2.4 says that $\mu_x * \mu_y$ is absolutely continuous exactly when some $g \in G$ satisfies

$$\mathcal{N}_x + \text{Ad}(g)\mathcal{N}_y = \mathfrak{g}.$$

Taking orthogonal complements turns this into

$$\mathfrak{z}_x \cap \text{Ad}(g)\mathfrak{z}_y = \{0\}. \quad (1)$$

Proof intuition

The $C_2 \times C_1$ centralizer is precisely the algebra of skew-Hermitian operators preserving one quaternionic line W . After an arbitrary conjugation, the other centralizer is diagonal in three quaternionic lines; its first two diagonal entries may point along any imaginary quaternion, while the third is restricted to one imaginary axis. Given the coordinates of a generator of W , choose a common right eigenvalue q so that the third diagonal entry lies on that axis, and conjugate q separately by the first two coordinates. This always manufactures a nonzero diagonal operator that preserves W , so (1) can never hold.

Main result

Theorem 1. *Let $x, y \in \text{Sp}(3)$ have types $C_2 \times C_1$ and $C_1 \times C_1$, respectively. Then*

$$\mu_x * \mu_y$$

is purely singular with respect to Haar measure. More sharply, the product of conjugacy classes $C_x C_y$ has exact real dimension 20 inside the 21-dimensional group $\text{Sp}(3)$. The same conclusion holds in every finite central quotient of $\text{Sp}(3)$.

Proof. Conjugate x to the displayed normal form and put $W = e_3\mathbb{H}$. Fix arbitrary $g \in G$, let $v_j = ge_j$, and write a nonzero generator $w = e_3$ of W as

$$w = v_1 r_1 + v_2 r_2 + v_3 r_3, \quad r_j \in \mathbb{H}.$$

Relative to v_1, v_2, v_3 , the algebra $\text{Ad}(g)\mathfrak{z}_y$ consists of diagonal operators with

$$a_1, a_2 \in \text{Im } \mathbb{H}, \quad a_3 \in \mathbb{R}i. \quad (2)$$

Suppose first that $r_3 \neq 0$. Define

$$q = r_3^{-1} i r_3, \quad a_3 = i, \quad a_j = \begin{cases} r_j q r_j^{-1}, & r_j \neq 0, \\ 0, & r_j = 0 \end{cases} \quad (j = 1, 2).$$

Quaternionic conjugation preserves $\text{Im } \mathbb{H}$, so (2) holds. Moreover $a_j r_j = r_j q$ for every j . Hence the corresponding diagonal operator T is nonzero and

$$Tw = wq \in W. \quad (3)$$

If $r_3 = 0$, choose any nonzero $q \in \text{Im } \mathbb{H}$, put $a_3 = 0$, and use the same formula for a_1, a_2 . At least one of r_1, r_2 is nonzero, so again $T \neq 0$ and (3) holds.

In both cases $T \in \text{Ad}(g)\mathfrak{z}_y$. Since T is skew-Hermitian and preserves W , it also preserves $W^\perp$: for $u \perp W$ and $w' \in W$,

$$\langle Tu, w' \rangle = -\langle u, Tw' \rangle = 0.$$

Thus T is block diagonal for $W^\perp \oplus W$, so $T \in \mathfrak{z}_x$. We have proved

$$\mathfrak{z}_x \cap \text{Ad}(g)\mathfrak{z}_y \neq \{0\} \quad \text{for every } g \in G.$$

Condition (1) always fails, and the source criterion proves singularity.

It remains to prove the sharp dimension assertion. Take the real orthogonal, hence quaternionic unitary, matrix

$$O = \frac{1}{9} \begin{pmatrix} 7 & -4 & -4 \\ -4 & 1 & -8 \\ -4 & -8 & 1 \end{pmatrix}.$$

All three coordinates $r_j = O_{3j}$ of e_3 in the column basis of O are nonzero real numbers. If $T \in \mathfrak{z}_x \cap \text{Ad}(O)\mathfrak{z}_y$, preservation of W gives $Tw = wq$. Comparing the three nonzero real coordinates yields $a_1 = a_2 = a_3 = q$; since $a_3 \in \mathbb{R}i$, necessarily $q \in \mathbb{R}i$. Conversely these scalar choices work because O is real. Therefore

$$\mathfrak{z}_x \cap \text{Ad}(O)\mathfrak{z}_y = \mathbb{R}(iI),$$

so the minimum intersection dimension is exactly one. Since $\dim_{\mathbb{R}} \mathfrak{sp}(3) = 21$, the maximum differential rank of the conjugacy-class product map is $21 - 1 = 20$. It is attained at O ; the constant-rank theorem locally and the universal upper bound globally give $\dim(C_x C_y) = 20$. Finite central quotients preserve this local dimension and the interior/absolute-continuity criterion. $\square$

Verification

The included script `code/verify_c3_pair.py` performs two independent checks. First, it tests the quaternionic construction in 20,000 random coefficient triples, including purity of the a_j and every identity $a_j r_j = r_j q$. Second, it computes over the rationals that the six off-block constraints induced by O have rank six on the seven parameters of $\mathfrak{z}_y$, with kernel

$$(1, 0, 0, 1, 0, 0, 1),$$

confirming the one-dimensional intersection and rank 20.

Upgrade attempts and limitations

The first upgrade sharpened singularity to exact codimension one. A second focused attempt tried to propagate the same construction to the higher exceptional C_n families. It does not do so: in higher rank the smaller x -eigenspace is an m -plane rather than a quaternionic line, and preserving that plane imposes a matrix invariance condition not solved by the single right-eigenvalue construction. Dimension count permits zero intersection there. Accordingly, this packet does not claim the general C_n, D_n conjecture or the higher exceptional families.

Novelty check and review recommendation

A bounded search covered the run indexes, arXiv:2110.06346, exact searches for the displayed C_3 type pair and its absolute continuity, later orbital-measure work by the source author, and searches combining the terms exceptional, orbital measure, and C_3 . No later resolution of this base case was found. Novelty confidence is moderate, not exhaustive.

Human review should focus on the identification of the two quaternionic Lie centralizers and on the orthogonal-complement form of source Proposition 2.4. Once those standard identifications are accepted, the intersection construction and the sharpness computation are elementary.

Reference

K. E. Hare, *The absolute continuity of convolutions of orbital measures in $SO(2n + 1)$* , arXiv:2110.06346 (2021), especially Proposition 2.4 and the discussion preceding Conjecture 5.9.